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High Resolution Fluid Tracking Using Subsurface DNA Diagnostics in High Water Cut Oilfield of the Ordos Basin, China

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Abstract

High Resolution Fluid Tracking is conductive to guiding the water shutoff operations in area with high water cut and complex residual oil distribution. Currently, chemical tracer technology is widely used, but can only be applied to a few wells due to its high cost and long monitoring cycle (e.g., only 10% of water shutoff wells in the study area). The rest rely on manual analysis from production history, leading to insufficient accuracy. Thus, subsurface DNA diagnostics, a new reservoir characterization tool, provides oil and gas operators with a high-resolution, non-invasive, cost-effective, and time-efficient method for measuring underground fluid movement. It has been applied in more than 200 wells in eight basins in North America and 38 wells in China. Subsurface DNA diagnostics is an integration of microbiology, geoscience and data analysis. Advances in the analysis of amplicon sequence datasets have introduced a methodological shift in how research teams investigate microbial biodiversity, away from sequence identity based clustering (producing Operational Taxonomic Units, OTUs) to denoising methods (producing amplicon sequence variants, ASVs). ASVs directly identify true biological sequences without requiring clustering thresholds, thereby enhancing interpretation accuracy and eliminating inconsistencies in artificial classification.

In this analysis, fluid samples from 7 production wells and 3 nearby injection wells were collected. Microbial DNA was extracted and sequenced to obtain Amplicon Sequence Variants (ASVs) that carry formation-specific signals of hydrocarbon fluid movement. The ASV dataset exhibited ultra-high dimensionality and extreme sparsity. We applied hierarchical clustering with automated parameter selection

through algorithm optimization. Using the zero-insensitive Bray-Curtis distance metric, clustering achieved a Silhouette Coefficient of 0.338 and Calinski-Harabasz Index of 2.525, confirming biologically meaningful microbial community grouping. Network graphs constructed with Cytoscape visualized well connectivity using the edge density theorem. Integration with chemical tracer monitoring validated the feasibility and accuracy of this subsurface DNA diagnostics technology. Overall, we significantly improved subsurface DNA diagnostics accuracy, minimized human errors in data processing, and optimized the high water cut oil fields' fluid movement monitoring plan, reducing costs by 60% and achieving direct economic benefits.

Introduction

Quantitative characterization of inter-well connectivity is crucial for selecting target formations and implementing effective injection-production adjustments or re-stimulation treatments. The traditional monitoring techniques, such as electrical log correlation, interference testing, geochemical analysis, tracer monitoring, and reservoir numerical simulation have limitations. Electrical log correlation and interference testing are highly subjective (Zhou et al., 2021; Kalule et al., 2024), reservoir numerical simulation offers broad coverage but low accuracy (Wong et al., 2020), geochemical analysis and tracer monitoring provide the highest accuracy but are costly, time-consuming, and potential environmental pollution (Sanni et al., 2018; Zhang et al., 2024). Therefore, there is an urgent need to develop a novel inter-well connectivity monitoring technology that is low-cost, environmentally friendly, and highly accurate.

Deep subsurface reservoir (exceeding 1 km depth) host unique ecosystems characterized by anaerobic, thermophilic (>80°C), and piezophilic (55.8 MPa) conditions, fostering distinct microbial communities exhibiting depth-dependent DNA variations and diverse subterranean signatures (Zhang et al., 2020; Zhang et al., 2022). Microbial DNA sequencing has been successfully applied within the petroleum technology—demonstrated in basins such as the Permian and Midland (USA) and in Japan—to monitor subsurface fluid flow pathways, confirming its technical feasibility (Peter et al., 2017; Silva et al., 2018;). Concurrently, the rapid advancement of DNA sequencing technology has drastically reduced costs (e.g., human whole-genome sequencing from \$1 million to \$100,000) (Mirza et al., 2024; Bayle et al., 2024). Consequently, integrating DNA sequencing with deep subsurface fluid flow analysis effectively overcomes the high cost and potential environmental disturbance associated with traditional methods.

The workflow for high-resolution fluid tracking using subsurface DNA diagnostics starts with DNA sequencing of fluid samples from injection and production wells. Next, shared microbial tracers are identified by analyzing microbial community composition using UPGMA clustering. A similarity matrix is then built by calculating Euclidean distances based on Operational Taxonomic Unit (OTU)-level species abundance data. Finally, Principal Coordinate Analysis (PCoA) visualizes microbial correlations in a lower-dimensional space to reveal subsurface flow pathways(Luke et al., 2019; Yang et al., 2025a; Yang et al., 2025b).

Microbiology has advanced significantly in recent years, with classification shifting from OTU to Amplicon Sequence Variants (ASV). OTUs are limited by artificially selecting cluster algorithms and arbitrary thresholds. In contrast, ASVs capture true single-nucleotide sequence variants without subjective thresholding. This provides a more accurate, precise, and standardized classification framework, greatly enhancing the resolution and interpretability of microbiological research.

In this analysis, fluid samples of Late Triassic period from 7 production wells and 3 nearby injection wells were collected. ASV-based diversity metrics were utilized to extract DNA characteristics for each sample, generating stratigraphically unique signals for hydrocarbon fluid movement. We used hierarchical clustering to process ASV-based dataset, and we also systematically compares distance metrics (Bray-Curtis, Jaccard, Euclidean, etc.) and linkage methods (Ward, complete, average, etc.) to automate optimal parameter selection. The Calinski-Harabasz index provides additional validation, ensuring biologically meaningful microbial community groupings. Then, network diagrams were constructed using Cytoscape to illustrate the connectivity between injection and production wells based on line density. The feasibility and accuracy of the Subsurface DNA diagnostic technology were verified in conjunction with the monitoring results of chemical tracers.

Sample and method

Sample

There are a total of 10 liquid samples from injection well and production well were collected from depths of 1800-2000 meters in the G area, which has been developed for 15 years. The area faces challenges including high water cut and uncertain remaining oil distribution. The oil-bearing area covers 44.83 km², with porosity and permeability of 8.6% and 0.38mD, classifying it as an ultra-low permeability reservoir. Figure 1 displays the well distribution, with red solid dots indicating production wells and blue solid dots indicating injection wells.

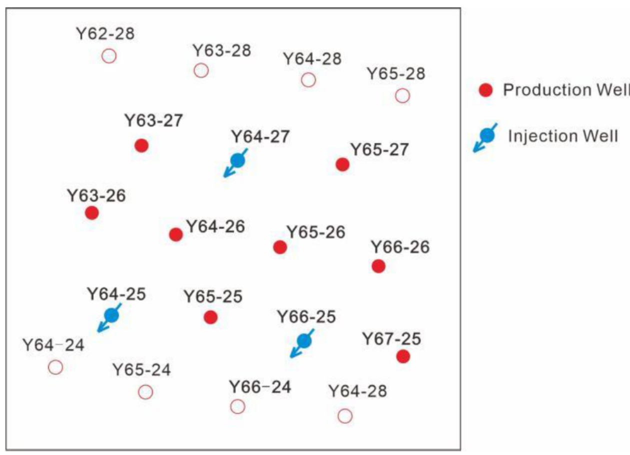

Figure 1—Well map of the study site.

Method

This methodology includes three key stages: sample collection and pretreatment, DNA extraction and sequencing, and the evaluation of fluid flow pathways based on ASV's set.

Sample collection and pretreatment. Fluid samples from production wells were collected directly at the wellhead, while injection well samples were obtained via valves at mixing stations. All samples were placed into sterile 30-50 mL conical flasks that had been thoroughly cleaned and UV-sterilized prior to use, with operators wearing sterilized gloves during collection to prevent contamination. Immediately after collection, sample flasks were immersed in cryopreservation solution containing ice packs to maintain on-site freezing and transported to the laboratory the same day. Samples were frozen promptly post-collection and remained sealed during transport/storage to minimize gas exchange. Upon laboratory arrival, samples were stored at -70 to -80°C for ≤1 week. For processing, samples were thawed slowly in an ice bath (avoiding ambient temperature exposure) with status checked every 10-15 minutes to ensure uniform thawing. Post-thaw, samples underwent enrichment centrifugation: microbial solutions were filtered through 0.45 μm and 0.22 μm membranes within a biosafety cabinet to concentrate microbial communities on filter surfaces. Using ethanol-sterilized instruments, membranes were sectioned and transferred to 50 mL centrifuge tubes containing 10 mL sterile phosphate buffered saline (PBS), followed by centrifugation at 14,000 rpm for 5 minutes. The resulting bacterial pellets were stored at -70 to -80°C for ≤1 week pending DNA extraction.

DNA extraction and sequencing. DNA was extracted from enriched and centrifuged samples for sequencing to analyze microbial community composition. The extraction involved: cell lysis by suspending samples in a specific buffer (containing Tris-HCl, EDTA, NaCl), chemical DNA extraction using proteinase and salts, and purification via ethanol precipitation (washed to remove salt impurities). This study targeted the V6 region of the 16S rRNA gene, amplified with primers U789F (TAGATACCBGGTAGC) and U1068R (CTGACGRCRRCCATGC) for PCR, enabling microbial diversity and taxonomic analysis. After sequencing, raw data underwent quality control prior to community composition analysis (Kobayashi et al., 2023).

Assessment of deep subsurface fluids flow paths based on ASV set. Initially, we employed a traditional workflow to assess inter-well connectivity. First, key microbial species were screened using UPGMA clustering based on species composition data, followed by PCoA of OTU data to evaluate shared species between injection and production wells (Sul et al., 2011; Werner et al., 2012). However, due to the impact of the low-permeability reservoir, this method revealed abundant shared species among the three injection well samples but failed to reveal shared species relationships between injection and production well samples. Consequently, we decided to return to the fundamental ASV set to re-analyze the data for a more accurate assessment of subsurface fluids flow paths. Meanwhile, we also conducted cluster analysis based on the microbial community composition for comparison.

Firstly, hierarchical clustering was performed to identify indicator microbial species in conjunction with community composition. To determine the optimal clustering solution, a double-loop approach was used to test combinations of multiple distance metrics (eg. Bray-Curtis, Jaccard, Euclidean, Manhattan, and correlation coefficient distances) and linkage methods (eg. Ward's method, furthest neighbor, average linkage, and weighted average linkage). Then, the silhouette coefficient was calculated for each combination, and the one with the highest score was selected. This enabled automatic matching of distance and linkage methods and the identification of the optimal clustering solution.

Importing ASV data into Cytoscape enables the construction of a sample-ASV bipartite network (nodes = samples/ASVs, edges = species presence relationships), directly identifying shared microbial species when multiple sample nodes (e.g., injection and production wells) connect to the same ASV node—indicating inter-well microbial transfer. This network quantifies inter-well connectivity strength by counting shared ASVs (edge connections) and utilizes topological parameters (e.g., centrality, modular clustering) to pinpoint key species and well groups with similar community structures. It overcomes the resolution limitations of traditional PCoA/UPGMA methods that obscure shared signals in low-permeability reservoirs. By analyzing microbial transfer pathways directly from high-resolution ASV data rather than relying on overall community similarity, this approach significantly enhances detection of weak connectivity signals amidst reservoir heterogeneity, providing a high-precision microbial evidence tool for assessing inter-well connectivity in low-permeability reservoirs.

Chemical tracer. To quantitatively characterize inter-well connectivity and compare it with subsurface DNA diagnostics results, the chemical tracer lanthanum (La) was injected into well Y64-27. This tracer was selected based on formation properties for its low background concentration and high stability. After injection into the target zone, surrounding production wells were sampled continuously at preset frequencies, and tracer concentrations in the produced fluid were analyzed using high-precision mass spectrometry. Subsequently, breakthrough time, front velocity, concentration profile shape, and partitioning ratio were analyzed. Combined with numerical simulation to invert parameters of high-permeability channels, this process quantitatively determined flow direction, connectivity degree, and sweep volume, thereby precisely characterizing inter-well connectivity.

Results

Microbial community composition of liquid samples

The microbial community composition based on DNA sequencing results is shown in Figure 2 (injection well samples are marked with boxes; others are production well samples). At the phylum level, all samples show low compositional variation. Proteobacteria and Firmicutes dominate, with mean relative abundances of 37.4% and 34.1%, respectively. Other phyla exhibit relative abundances ranging from 0.2% to 5% (Figure 2a).In 3 samples, Proteobacteria abundance higher than Firmicutes, indicating high-permeability seepage channels (Probandt et al., 2017). In the remaining 7 samples, higher Firmicutes abundance reflects diffusion-limited environments. At the class level, microbial community composition begins to show variation. Gammaproteobacteria, Clostridia, and Desulfotomaculia dominate, with mean relative abundances of 31.2%, 10.2%, and 9.4%, respectively. Other classes exhibit relative abundances ranging from 0.8% to 6% (Figure 2b). Notably, all three classes are anaerobic bacteria. Specifically, the enrichment of Clostridia and Desulfotomaculia further indicates exogenous organic matter invasion (Xing et al., 2023), suggesting microbial community composition can reflect the effectiveness of water injection operations.

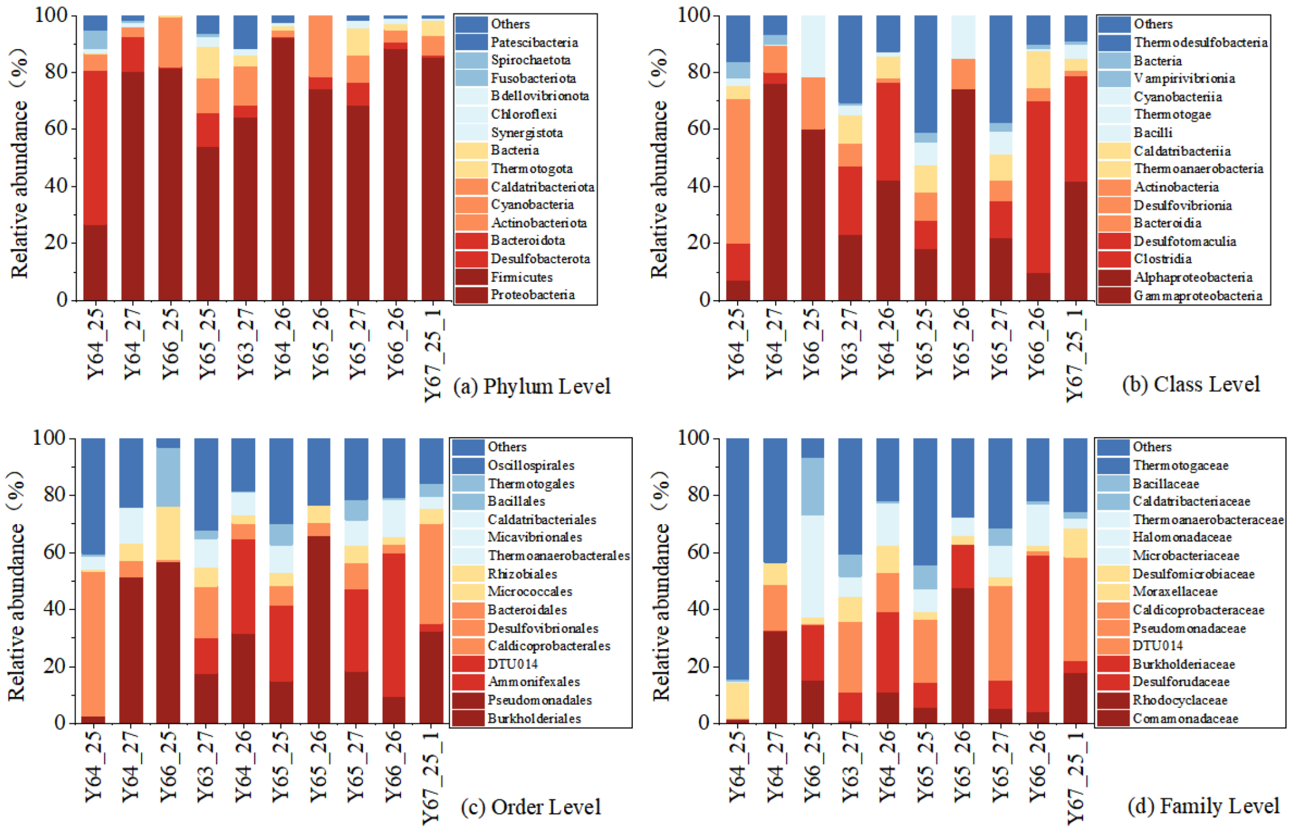

Figure 2—Stacked bar chart of the relative abundance percentage of microbial community composition from fluids samples by injection and production well at the (a) phylum, (b) class, (c) order and (d) family levels.

At the order level, microbial classification becomes more refined, providing clearer environmental indications. Burkholderiales dominates with a significantly higher mean relative abundance (21.4%), while other orders show minimal variation (1.3%-8.8%) (Figure 2c). The consistent prevalence of Burkholderiales across all samples suggests exogenous aerobic intrusion into facultative anaerobic conditions, confirming widespread effectiveness of water injection in most samples. At the family level, microbial communities exhibit low compositional variability, with average relative abundances ranging from 1.8% to 12%. Notably, Comamonadaceae, Desulforudaceae, Burkholderiaceae, DTU014, and Caldicoprobacteraceae show higher abundances (6.3%-12.2%) (Figure 2d). Comamonadaceae and Burkholderiaceae (facultative aerobes) serve as bioindicators for water flooding efficiency (Sharuddin et al., 2023) while the enrichment of strictly anaerobic Desulforudaceae and DTU014 (a Clostridia-related family) reflects unflooded anaerobic zones. Caldicoprobacteraceae indicates deep thermophilic conditions (Bouanane et al., 2011). Given the stronger environmental responsiveness at the family level, we selected these data for cluster analysis to compare with ASV-based results.

Shared microbial community between injection and production wells

Based on microbial composition analysis, we performed cluster analysis using family-level taxonomic profiles. These results were compared with ASV-based clustering to illustrate shared taxa between injection and production wells.

Model performance. Tables 1 and 2 show the clustering results for microbial community composition data and ASV set, respectively. Due to differences in their data characteristics, the optimal clustering methods vary. For microbial community composition data, which is lower in dimension and more evenly distributed, Euclidean distance combined with Complete linkage performed best. It achieved a Silhouette Score of 0.344 and a Calinski-Harabasz index of 5.717. This method works well with continuous numerical data and produces compact, spherical clusters.

Table 1—Clustering algorithm performance comparison based on relative abundance microbial community composition

Algorithm Combination	Silhouette	Calinski-Harabasz	Clusters
euclidean + complete	0.344	5.717	3
cityblock + complete	0.344	5.717	3
correlation + average	0.344	5.717	3
correlation + weighted	0.344	5.717	3
braycurtis + complete	0.34	6.352	2
braycurtis + average	0.34	6.352	2
braycurtis + weighted	0.34	6.352	2
euclidean + ward	0.34	6.352	2
correlation + complete	0.34	6.352	2
cityblock + weighted	0.305	5.343	4
euclidean + average	0.218	5.065	5
euclidean + weighted	0.218	5.065	5
cityblock + average	0.218	5.065	5

Table 2—Clustering algorithm performance comparison based on ASV set

Algorithm Combination	Silhouette	Calinski-Harabasz	Clusters
braycurtis + complete	0.338	2.525	2
braycurtis + average	0.338	2.525	2
braycurtis + weighted	0.338	2.525	2
euclidean + ward	0.338	2.525	2
euclidean + complete	0.182	1.747	8
euclidean + average	0.182	1.747	8
euclidean + weighted	0.182	1.747	8
correlation + average	0.182	1.747	8
correlation + weighted	0.182	1.747	8
cityblock + complete	0.117	1.969	6
correlation + complete	0.117	1.969	7
cityblock + average	0.062	2.099	7
cityblock + weighted	0.062	2.099	7

In contrast, ASV set is high-dimensional, sparse (with many zeros), and compositional. For this type of data, Bray-Curtis distance with Complete linkage was optimal, yielding a Silhouette Score of 0.338 and a Calinski-Harabasz index of 2.525. Bray-Curtis distance is designed for ecological data and ignores many zeros—focusing only on abundance differences of species present in samples. This better aligns with microbial community analysis and reduces noise from sparse, high-dimensional data. Therefore, the choice of clustering method depends on data structure: Euclidean distance suits moderate-dimensional, uniform data, while Bray-Curtis is better for high-dimensional, sparse compositional data like ASVs.

Shared microbial community. The clustering analysis generated dendrograms based on microbial community composition and ASVs, relatively. Samples sharing shorter branch lengths and residing on the same branches demonstrated higher similarity in microbial community composition. At the family level, fluid samples clustered into three groups. Group one comprised injection wells Y64-27 and Y64-25. Group two included production well Y65-26 and injection well Y66-25. The remaining samples formed the third group. Figure 3a reveals short branch lengths between injection well Y66-25 and production well Y65-26, indicating they shared more microbial community. Similarly, short branch lengths existed between injection wells Y64-27 and Y64-25, also suggesting significant shared microbial community. Collectively, the four samples Y64-27, Y64-25, Y65-26, and Y66-25 exhibited relatively short branch lengths, demonstrating greater overall taxonomic similarity among them.

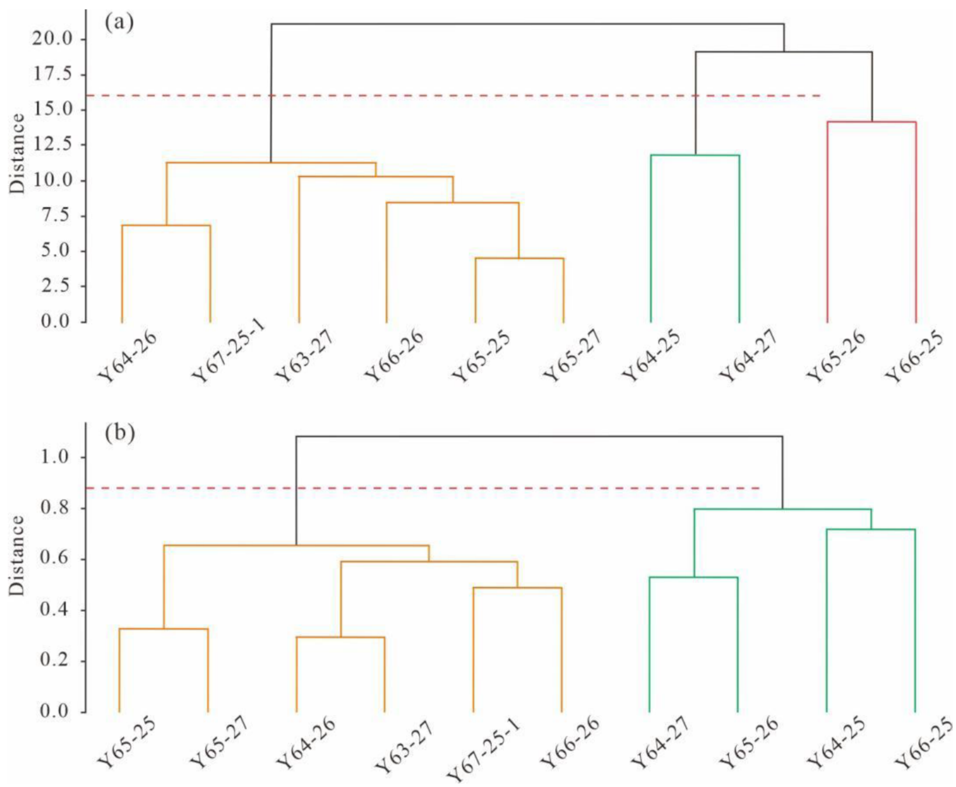

Figure 3—Hierarchical clustering dendrogram based on (a) microbial community composition and (b) ASVs.

Figure 3b displays the ASV-based dendrogram where samples cluster into two groups. Three injection wells and production well Y65-26 form one group, while the remaining fluid samples constitute the other group. This clustering pattern provides valuable insights for assessing inter-well connectivity. Specifically, the short branch length between injection wells Y64-25 and Y66-25 indicates greater shared microbial taxa, resulting from their shared injection pipeline. However, although injection well Y64-27 belongs to the same group as Y64-25 and Y66-25, it originates from a different injection pipeline, resulting in lower ASV similarity. In contrast, stronger connectivity is observed between injection well Y64-27 and production well Y65-26.

Fluids flow pathways between injection and production wells

We imported ASV data from 10 fluid samples into Cytoscape software, which analyzed shared microbial taxa by constructing an ASV-sample bipartite network model. This model treats ASV nodes and sample nodes as two distinct entities, establishing a connection edge between them whenever an ASV is detected in a sample (abundance > 0). The model directly reveals sharing relationships where ASV nodes with high node degree (connected to many samples) represent core shared taxa, while closely clustered sample node groups indicate similar microbial community composition. By performing node degree analysis to quantify the number of samples connected to each ASV, researchers can clearly identify sample clusters exhibiting high proportions of shared taxa, thus providing insights into inter-well fluid connectivity.

As shown in Figure 4, injection wells (blue) and production wells (red) form well group connectivity through shared ASVs (green). Well-to-well connectivity is quantified by calculating the proportion of connection edges between injection and production wells relative to the total edges within each well group. Results indicate significantly higher shared ASVs and stronger connectivity in Y64-27 and Y64-25 well groups, whereas Y66-25 exhibits fewer shared ASVs with some wells showing closer associations to other groups. Detailed quantification reveals 42 core shared ASVs with 101 total edges between injection well Y64-27 and production wells Y63-27, Y65-27, and Y65-26, where Y65-26 accounts for the highest proportion (29 edges, 28.7%), followed by Y65-27 (32 edges, 31.6%), Y63-27 (33 edges, 32.7%), and Y64-25 (7 edges, 7.0%). The Y64-25 well group contains 45 core shared ASVs with 71 total edges, primarily distributed in Y64-26 (28 edges, 39.4%) and Y65-25 (32 edges, 45.0%), while other wells collectively contribute 11 edges (15.5%).

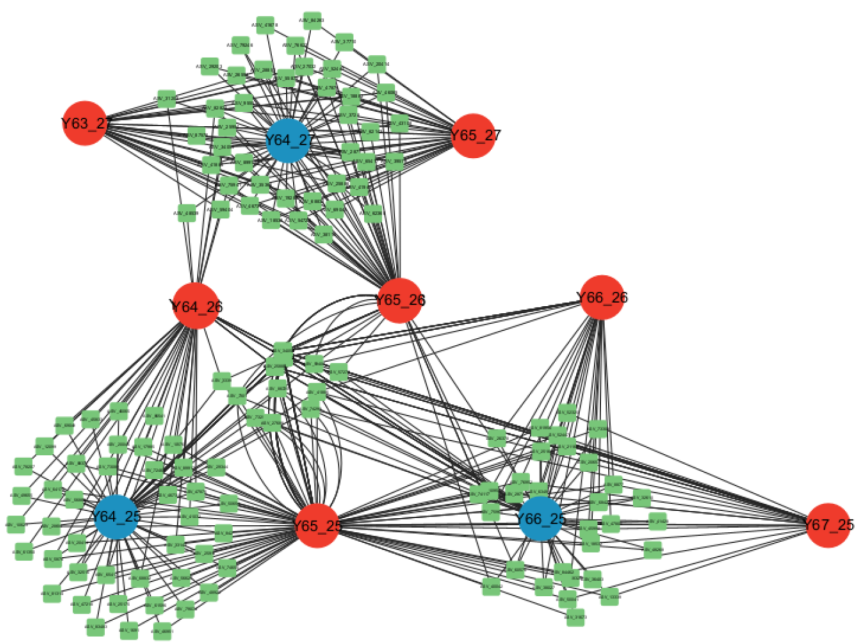

Figure 4—Bipartite network representation of ASV data from 10 subsurface fluid samples constructed with Cytoscape (Blue is injection well, red is production well and green is shared ASVs).

Chemical tracer

The Y64-27 well group micro-tracer monitoring test was conducted to quantitatively characterize interwell connectivity. First, the tracer was selected, choosing lanthanum for its low background concentration and environmental friendliness. The tracer dosage was determined as 15.3 kg using the maximum dilution volume method, with a safety factor of 200, and a prepared concentration of 1%. It was injected into the injection well. The next day, systematic sampling commenced from eight surrounding production wells, lasting 82 days. Oil samples were collected at an initial frequency of every other day, gradually decreasing over time. After oil-water separation and filtration, lanthanum concentration was detected, achieving a detection limit of 0.002 μg/L. The data was analyzed using a one-dimensional convection-diffusion model incorporating adsorption hysteresis and a streamline model to quantitatively characterize interwell parameters.

The tracer concentration curve and rose plot are shown in Figure 5. Tracer was detected in only three wells, including Y63-27, Y65-26, and Y65-27. Breakthrough times were 12 days, 14 days, and 16 days respectively. Peak concentrations measured 23.2 ng/mL, 19.2 ng/mL, and 22.4 ng/mL respectively. The water front velocity ranged from 17.67 to 21.95 m/day with an average of 19.45 m/day. A velocity ratio of 1.13 indicated good interwell channel homogeneity. Injected water allocation showed tracer-responding wells dominated the flow distribution with Y63-27 accounting for 33.5%, Y65-26 accounting for 23.1%, and Y65-27 accounting for 43.4% (Figure 6). However, actual water production was lower than the allocated volume, suggesting undetected flow paths or formation retention. These results provide a basis for identifying dominant waterflooding directions quantitatively.

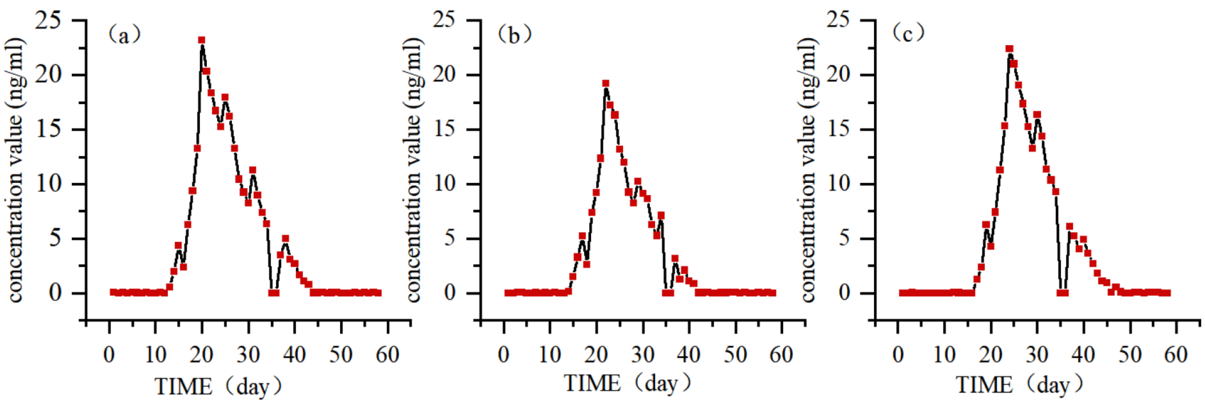

Figure 5—The tracer concentration curve distribution diagram of Well (a) Y63-27, (b) Y65-26 and (c) Y65-27.

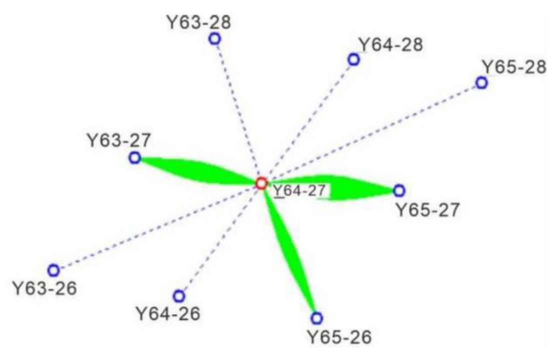

Figure 6—Rose diagram of chemical tracer for Y64-27 well group.

Discussion

Clustering Analysis Comparison between ASV and Microbial Composition data

ASV-based clustering analysis revealed high similarity between injection well Y64-27 and production well Y65-26, indicating good connectivity. At the phylum, order, and family levels, their microbial compositions showed strong consistency, both dominated by Gammaproteobacteria and Alphaproteobacteria at the phylum level, signifying injection response. Additionally, liquid samples from injection well Y64-25 and production well Y66-25 are cyclic reinjection of water in the field, also demonstrated high clustering consistency matching objective conditions. However, family-level clustering indicated high similarity between injection wells Y64-27 and Y64-25, contradicting reality as they belong to different water injection lines: Y64-27 injects freshwater while Y64-25 re-injects produced water. Thus, ASV-based clustering more accurately reflects shared microbial populations between injection and production wells.

Comparison between bipartite network from ASVs and chemical tracers

ASVs Bipartite network and chemical tracers are used to analysis inter-well connectivity quantitatively. The result shows the significant connectivity between injection well Y64-27 and production wells Y63-27 and Y65-27. ASV analysis shows the highest shared ASV count between Y64-27 and Y63-27 (33 ASVs) and 32 shared ASVs with Y65-27, indicating similar microbial community composition. Chemical tracer data confirm that the Y63-27 exhibited the earliest tracer breakthrough (Day 12) with an initial concentration of 0.56 ng/mL and a peak of 23.2 ng/mL, while Y65-27 showed similarly rapid response, verifying their strong hydrodynamic connection to the injection well. The strong correlation between microbial community composition and tracer transport patterns confirms multi-data consistency in reservoir connectivity assessment, validating the feasibility and accuracy of subsurface DNA diagnose for evaluating inter-well connectivity.

Conclusion

1. Processing ASV data with an optimized analytical algorithm (Bray-Curtis distance + complete linkage) significantly enhances microbial tracing resolution. In the Ordos Basin high-water-cut oilfield application, this method achieved a silhouette coefficient of 0.338 and a Calinski-Harabasz index of 2.525. Combined with bipartite network analysis, it precisely identifies inter-well fluid pathways (e.g., 33 shared ASVs between wells Y64-27 and Y65-26, representing 32.7% connectivity strength). This overcomes the resolution limitations of traditional OTU methods in tight reservoirs, enabling high-resolution inter-well connectivity monitoring.
2. Subsurface DNA diagnostics and chemical tracer monitoring data show strong correlation. High similarity of ASVs between injection well Y64-27 and production wells Y63-27/Y65-27 (up to 65 shared ASVs) correlates with tracer breakthrough times (12–16 days) and peak concentrations (19.2–23.2 ng/mL). This dual-validation approach confirms that shared microbial community composition quantitatively characterize dominant flow channels, providing a novel analytical tool for reservoir inter-well connectivity.
3. Compared with chemical tracer methods, which have high costs due to the need for prolonged injection of large volumes of chemical reagents, subsurface DNA diagnostics involves only the expense of DNA sequencing of fluid samples, significantly reducing overall expenditures. This technology simultaneously eliminates the potential environmental risks compared with chemical tracers. By utilizing hierarchical clustering algorithms to drive automated iterative parameter optimization, the approach minimizes subject intervention errors, providing an efficient method for evaluating inter-well connectivity in high-water-cut oilfields.

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